30 The atomic radii and the first ionisation energies of the metals iron and cobalt are very similar, despite the fact that cobalt has a higher nuclear charge than iron.

Which statement explains one of these similarities?

A The atomic radius of Co is similar to that of Fe because the two 4s electrons cause the same shielding in both atoms.

B The atomic radius of Co is similar to that of Fe because Co has an extra 4p electron which increases the shielding.

C The first ionisation energy of Co is similar to that of Fe because the 3d electron being removed is shielded by two 4s electrons in both atoms.

D The first ionisation energy of Co is similar to that of Fe because the 4s electron being removed is shielded by an extra 3d electron in Co.